

Task-01: Visualizing the Distribution of Population Data Using Bar Chart and Histogram

Objective

The objective of this task is to visualize the distribution of a continuous or categorical variable using appropriate graphical techniques. In this analysis, the **World Bank Population Dataset** was used to explore the distribution of country populations in the year **2023**. Two visualization techniques were applied:

- Histogram (Continuous Variable)
 - Bar Chart (Categorical Variable)
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Dataset Description

The dataset was obtained from the **World Bank Open Data** and contains annual total population data for countries and regions worldwide from **1960 to 2025**.

Dataset Information

- Dataset Name: World Bank Population Dataset
- Indicator: Population, Total
- Year Selected: 2023
- Variable Used: Total Population

After loading the dataset, the first four metadata rows were skipped to obtain the actual data.

```
df = pd.read_csv(

"population_data/API_SP.POP.TOTL_DS2_EN_csv_v2_3107
.csv",

    skiprows=4

)
```

[4]:

	Country Name	Country Code	Indicator Name	Indicator Code	1960	1961	1962	1963	1964	1965	...	2017	2018	2019
0	Aruba	ABW	Population, total	SP.POP.TOTL	54922.0	55578.0	56320.0	57002.0	57619.0	58190.0	...	108735.0	108908.0	109200.0
1	Africa Eastern and Southern	AFE	Population, total	SP.POP.TOTL	130075728.0	133534923.0	137171659.0	140945536.0	144904094.0	149033472.0	...	640058741.0	657801085.0	675950100.0
2	Afghanistan	AFG	Population, total	SP.POP.TOTL	9035043.0	9214083.0	9404406.0	9604487.0	9814318.0	10036008.0	...	35688935.0	36743039.0	37856100.0
3	Africa Western and Central	AFW	Population, total	SP.POP.TOTL	97630925.0	99706674.0	101854756.0	104089175.0	106384410.0	108754449.0	...	440150152.0	451395343.0	462522200.0
4	Angola	AGO	Population, total	SP.POP.TOTL	5231654.0	5301583.0	5354310.0	5408320.0	5464187.0	5521981.0	...	30234839.0	31297155.0	32375600.0

5 rows × 15 columns

Data Preparation

The population values for the year **2023** were extracted, and missing values were removed.

```
year = "2023"
```

```
population = df[year].dropna()
```

Since country populations vary from a few thousand to several billion, a logarithmic transformation (Log10) was applied to improve the visualization.

Visualization 1: Histogram

Purpose

A histogram was created to examine the distribution of country populations across the world.

Code

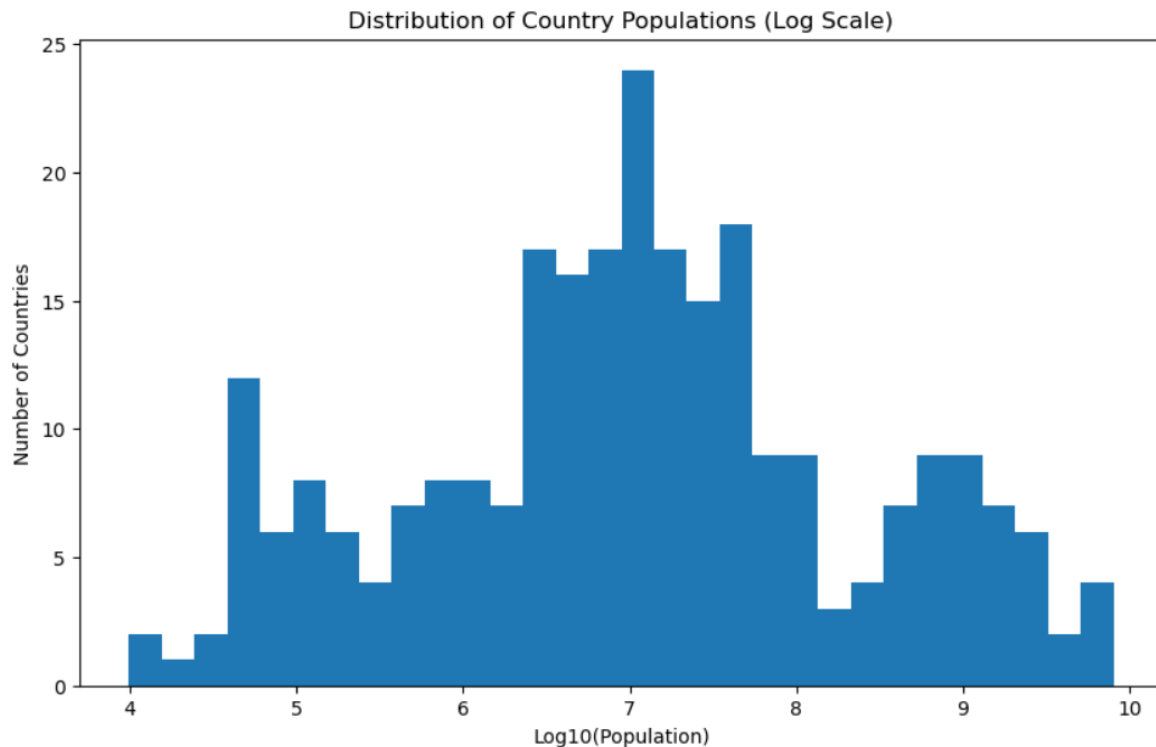
```
import numpy as np
import matplotlib.pyplot as plt

plt.figure(figsize=(10,6))

plt.hist(np.log10(population), bins=30)

plt.title("Distribution of Country Populations (Log Scale)")
plt.xlabel("Log10(Population)")
plt.ylabel("Number of Countries")

plt.show()
```



Interpretation

The histogram displays the distribution of populations after applying a logarithmic transformation.

Observations

- Most countries fall within the middle population ranges.
- Very few countries have extremely small populations.
- A small number of countries have exceptionally large populations.
- The logarithmic scale makes the distribution easier to understand by reducing the effect of extreme values.

Conclusion

The distribution is positively skewed in its original form because only a few countries have populations exceeding hundreds of millions, while most countries have much smaller populations.

Visualization 2: Bar Chart

Purpose

A bar chart was created to compare the countries (or regional groups) with the largest populations in 2023.

Code

```
top10 = df[["Country Name", year]].dropna()
```

```
top10 = top10.sort_values(by=year,  
ascending=False).head(10)
```

```
plt.figure(figsize=(12,6))
```

```
plt.bar(top10["Country Name"], top10[year])
```

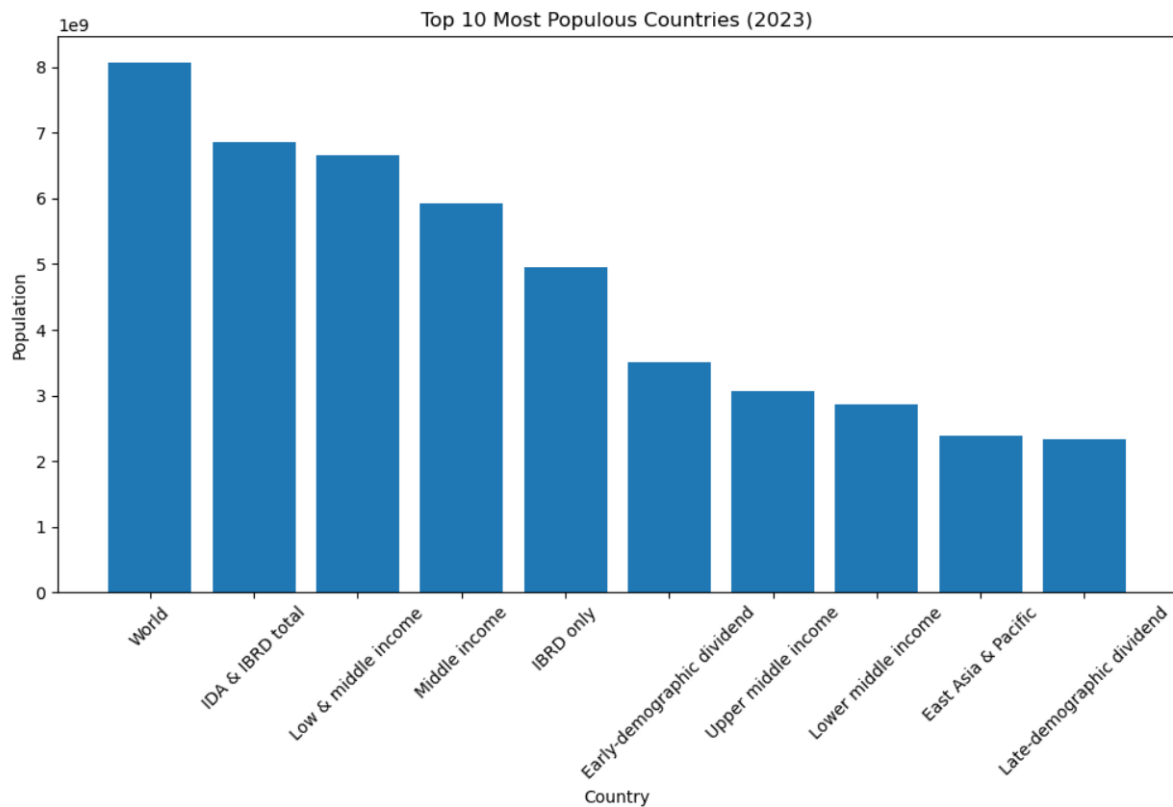
```
plt.xticks(rotation=45)
```

```
plt.title("Top 10 Most Populous Countries (2023)")
```

```
plt.xlabel("Country")
```

```
plt.ylabel("Population")
```

```
plt.show()
```



Horizontal Bar Chart

To improve readability, a horizontal bar chart was also created.

```
plt.figure(figsize=(10,6))
```

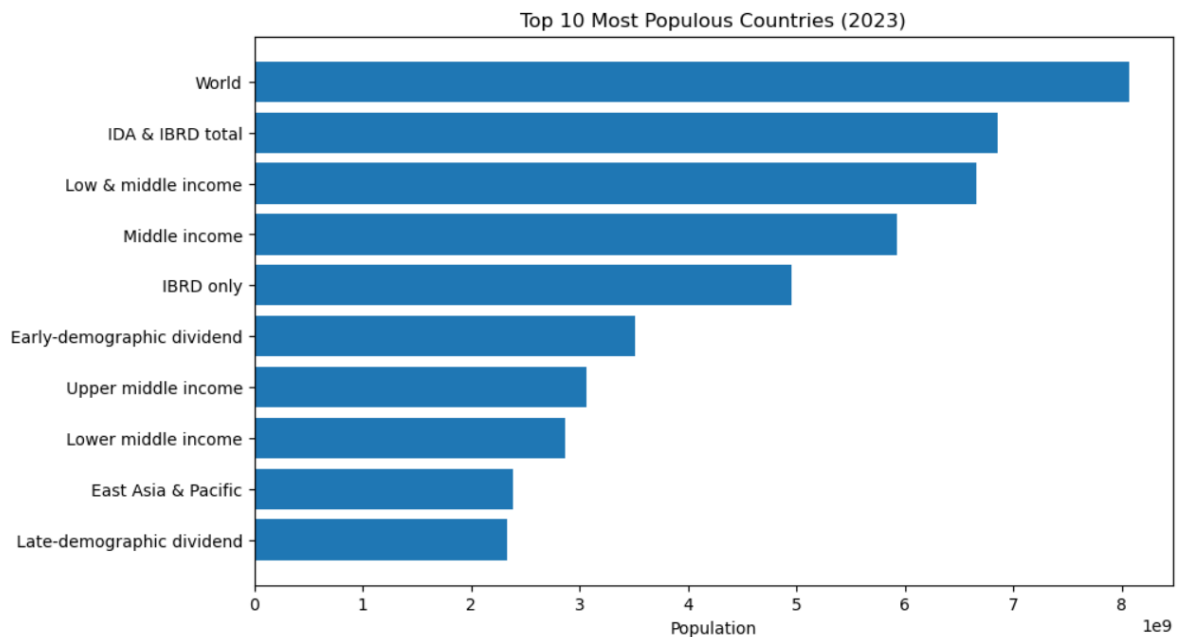
```
plt.barh(top10["Country Name"], top10[year])
```

```
plt.title("Top 10 Most Populous Countries (2023)")
```

```
plt.xlabel("Population")
```

```
plt.gca().invert_yaxis()
```

```
plt.show()
```



Interpretation of the Bar Chart

The bar chart compares the population sizes of the ten largest entries in the dataset.

Observations

- **World** has the highest total population because it represents the global population.
- **IDA & IBRD Total** and **Low- & Middle-Income** groups also have very large populations since they combine multiple countries.
- **East Asia & Pacific** is among the largest regional groups due to countries such as China and Indonesia.
- Population sizes vary considerably across different regions and income groups.

Note:

The World Bank dataset includes **aggregated regions and income groups**, not only individual countries. Therefore, entries such as **World, Middle Income, and East Asia & Pacific** appear among the largest populations.

Overall Findings

The analysis provides important insights into the distribution of global population data.

- The histogram shows that population data are highly uneven and positively skewed.
 - Most countries have relatively small populations compared with a few highly populated countries.
 - The logarithmic transformation improves the visibility of the distribution.
 - The bar chart clearly compares the largest population groups and highlights the significant differences among them.
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Conclusion

This task successfully demonstrated how visualization techniques can be used to understand the distribution of population data.

The histogram effectively illustrated the distribution of a continuous variable (population), while the bar chart provided a clear comparison of the largest population groups. These visualizations help identify trends, detect skewness in the

data, and compare population sizes across countries and regions.

Overall, data visualization is an essential step in exploratory data analysis because it transforms raw numerical data into meaningful insights that are easier to interpret and communicate.